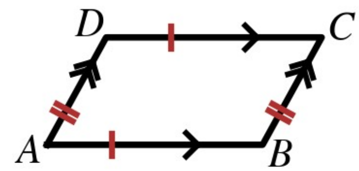
Speak **L**ike **A** **M**athematician:



quadrilateral:

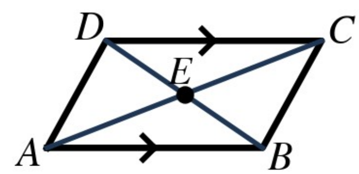
parallelogram:

diagonals of a quadrilateral:

Properties of Parallelograms

If a quadrilateral is a parallelogram, then: opposite sides are ____________________ · opposite angles are ____________________ · consecutive angles are ____________________ · the diagonals ____________________ each other.

Example 1 – Prove Opposite Sides Are Congruent



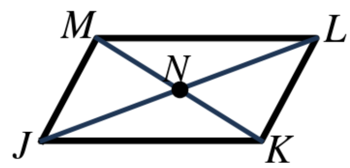
Given: ABCD is a parallelogram with diagonal $\overline{AC}$

Prove: $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$

Statements	Reasons
1. ABCD is a parallelogram with diagonal $\overline{AC}$	**1.** Given
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**
6.	**6.**

Example 2 – Use the Properties

In ▱JKLM at the right, the diagonals meet at N. KL = 58 mm, KN = 26 mm, and m∠MJK = 51°.



a) Find MJ.

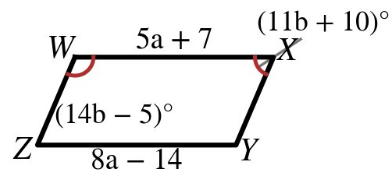
b) Find m∠JML.

c) Find m∠KLM.

d) Find KM.

Example 3 – Solve for the Variables

WXYZ at the right is a parallelogram.



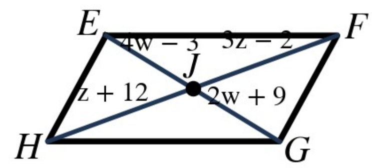
a) Find a, then find ZY.

b) Find b, then find m∠X.

c) Find m∠Z.

Example 4 – Diagonals

In ▱EFGH at the right, the diagonals meet at J. EJ = 4w − 3, JG = 2w + 9, FJ = 3z − 2, and JH = z + 12.



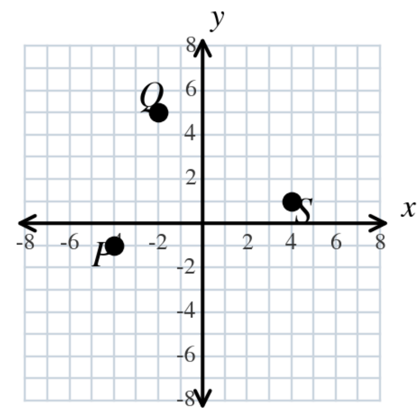
a) Find w.

b) Find z.

c) Find EG and FH.

Example 5 – Find the Fourth Vertex

Three vertices of ▱PQRS are P(−4, −1), Q(−2, 5), and S(4, 1). Find vertex R.



a) Which segment must be parallel and congruent to $\overline{PQ}$?

b) Use the 'same rise, same run' idea to find R.

c) Check with the diagonals.